

Sets and Sequences*

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1 Sets

A **set** is characterized by what objects it contains and we call them its **elements** or **members**.

- $x \in X$: the object x **belongs to**, or **is an element of** the set X .
- $X = \{1, \pi, \text{apple}, \text{economics}\}$.
- $X = \{n \in \mathbb{Z} : n^2 = 4\}$.
- $\emptyset$: **empty set**.

1.1 Inclusion

We say that A is **included in** B , or that A is a **subset of** B and denote as $A \subseteq B$ (or $A \subset B$) if every member of A is a member of B : $x \in A \Rightarrow x \in B$. Any set is a subset of itself.

- $A = B \Leftrightarrow A \subseteq B$ and $B \subseteq A$.
- $A \subsetneq B \Leftrightarrow A \subset B$ and $B \subset A$ is not equal. In this case, we say that A is a **proper subset** of B .
- The inclusion is **transitive**: if $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.
- The set of all subset of a set X is called the **power set** of X and note as $P(X)$ or 2^X .

*This material is largely based on the lecture notes of math camp in Arizona University and Columbia University.

1.2 Union and Intersection

The **intersection** of two sets A and B is the set of all elements that belong to both A and B :

$$A \cap B := \{x : x \in A \text{ and } x \in B\}$$

- If no element lies in both A and B , $A \cap B$ is still defined. It is defined to be the empty set, $A \cap B = \emptyset$. We say that A and B are **disjoint**.
- The intersection is commutative: $A \cap B = B \cap A$.
- The intersection is associative: $(A \cap B) \cap C = A \cap (B \cap C)$.
- $A \cap B = A \Leftrightarrow A \subseteq B$.

The **union** of two sets A and B , noted $A \cup B$ is the set of all elements that belong to either A or B (possibly both):

$$A \cup B := \{x : x \in A \text{ or } x \in B\}$$

- The union is commutative: $A \cup B = B \cup A$.
- The union is associative: $(A \cup B) \cup C = A \cup (B \cup C)$.
- $A \cup B = A \Leftrightarrow B \subseteq A$.

More generally, we can talk about the union of any collection of sets, possibly infinite. Let Θ be a set (potentially infinite) and let $\{A_\theta\}_{\theta \in \Theta}$ be a collection of sets indexed by $\theta \in \Theta$. That is, for each $\theta \in \Theta$, A_θ is a set in the collection. There are as many sets in the collection $\{A_\theta\}_{\theta \in \Theta}$ as there are elements in Θ . Generalize the notation we defined previously to allow infinite union/intersection:

$$\bigcup_{\theta \in \Theta} A_\theta := \{x : \exists \theta \in \Theta \text{ s.t. } x \in A_\theta\}$$
$$\bigcap_{\theta \in \Theta} A_\theta := \{x : x \in A_\theta, \forall \theta \in \Theta\}$$

Notice that when the index set Θ has only two elements θ_1 and θ_2 , the notations above reduces to pairwise union/intersection we defined at the beginning.

- The intersection is distributive w.r.t. the union:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \text{ and}$$

$$B \cap \left(\bigcup_{\theta \in \Theta} A_\theta \right) = \bigcup_{\theta \in \Theta} (B \cap A_\theta).$$

- The union is distributive w.r.t. the intersection:

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C) \text{ and}$$

$$B \cup \left(\bigcap_{\theta \in \Theta} A_\theta \right) = \bigcap_{\theta \in \Theta} (B \cup A_\theta).$$

1.3 Difference and complement

Given two sets A and B , the **set difference** $A - B$ or $A \setminus B$ is the set of all x that belong to A but not to B :

$$B \setminus A := \{x \in B : x \notin A\}$$

Notice that in the definition above A does not need to be a subset of B .

Let U be the set of all objects that are relevant to the current question and A be a subset of U . Define the **complement** of A as

$$A^c := U \setminus A$$

- Notice that the notation A^c makes sense only when the “universe” U has no ambiguity in the question. Whenever U currently used is ambiguous, we should be explicit about U by writing $U \setminus A$ instead of A^c .
- Note $x \notin A$ for $x \in A^c$ ($x \in U$ implicitly).
- $(A^c)^c = A$.

Proposition 1.1 (De Morgan’s Law). *Let $\{A_\theta\}_{\theta \in \Theta}$ be a collection of subsets of U . We have*

$$(1) \left(\bigcup_{\theta \in \Theta} A_\theta \right)^c = \bigcap_{\theta \in \Theta} A_\theta^c.$$

$$(2) \left(\bigcap_{\theta \in \Theta} A_\theta \right)^c = \bigcup_{\theta \in \Theta} A_\theta^c.$$

1.4 Cartesian Product

Define the **Cartesian product** of set A and B as

$$A \times B := \{(a, b) : a \in A \text{ and } b \in B\}$$

The notation (a, b) is for an ordered pair. The pair is ordered in the sense that (a, b) is different from (b, a) . Cartesian product of A and B is the set of all ordered pairs whose first element is taken from A and whose second element is taken from B . Notice that $A \times B \neq B \times A$. We can also talk about the Cartesian produce of multiple sets $A_1 \times A_2 \times \cdots \times A_n$ or simply $\prod_{i=1}^n A_i$, defined as

$$A_1 \times A_2 \times \cdots \times A_n := \{(a_1, a_2, \cdots, a_n) : a_i \in A_i, \forall n = 1, 2, \cdots, n\}$$

If all A_i 's are the same, order does not matter and we can denote as $A \times A \times \cdots \times A$ as A^n .

1.5 Addition of Sets

The sum of sets is an important concept in Economics. For example, if a firm has a production process by which it can achieve any input-output n -tuple in the set $X_1 \subseteq \mathbb{R}^n$ and also another process by which it can achieve any n -tuple in $X_2 \subseteq \mathbb{R}^n$, then we might assume that altogether it could achieve any vector $\mathbf{z} = \mathbf{x}_1 + \mathbf{x}_2$ such that $\mathbf{x}_1 \in X_1$ and $\mathbf{x}_2 \in X_2$ (if operating one of the processes imposes no external effects on the other process). Or if there are m firms and Firm i can achieve any input-output plan $\mathbf{x}_i \in X_i \subseteq \mathbb{R}^n$ ($i = 1, \cdots, m$), then we might expect the economy to be able to achieve, in the aggregated, any input-output vector $\mathbf{z} \in \mathbb{R}^n$ that is the sum $\mathbf{x}_1 + \cdots + \mathbf{x}_m$ of vectors $\mathbf{x}_i$ that each lie in the respective sets X_i (if a firm's production imposes no external effects on the other firms' production possibilities). This motivates the following definition:

Definition 1.2. *Let X_1 and X_2 be subsets of a vector space V . The sum $X_1 + X_2$ is defined as follows:*

$$X_1 + X_2 = \{\mathbf{x}_1 + \mathbf{x}_2 | \mathbf{x}_1 \in X_1 \text{ and } \mathbf{x}_2 \in X_2\}.$$

For sets $X_1, \cdots, X_m \subseteq V$ we define $X_1 + \cdots + X_m$ or $\sum_{i=1}^m X_i$, similarly:

$$X_1 + \cdots + X_m = \{\mathbf{x}_1 + \cdots + \mathbf{x}_m | \mathbf{x}_i \in X_i, i = 1, \cdots, m\}$$

For intuition, it is helpful to note that we can write the definition equivalently as

$$X_1 + X_2 = \{\mathbf{z} \in V \mid \exists \mathbf{x}_1 \in X_1 \& \mathbf{x}_2 \in X_2 \text{ such that } \mathbf{x}_1 + \mathbf{x}_2 = \mathbf{z}\}$$

If one of the sets is a singleton - for example, $A = \{x\}$ - we typically write $\mathbf{x} + B$ instead of $\{x\} + B$ or $A + B$.

Example 1: $\mathbf{0} + A = A$ for any set A .

Example 2: $A = \{\mathbf{x} \in \mathbb{R}^2 \mid x_1^2 + x_2^2 \leq 1\}$, $B = \{(2, 2)\}$.

Example 3: $A = \{\mathbf{x} \in \mathbb{R}^2 \mid x_1^2 + x_2^2 \leq 1\}$, $B = \{(2, 2), (2, 1)\}$.

Example 4: $A = \{\mathbf{x} \in \mathbb{R}^2 \mid 1 \leq x_1 \leq 2, 1 \leq x_2 \leq 3\}$, $B = \{\mathbf{x} \in \mathbb{R}^2 \mid (x_1 - 4)^2 + (x_2 - 3)^2 \leq 1\}$.

Example 5: Suppose one firm's production possibilities are described by the production function f_A and the associated inequality $y \leq f_A(x) = \frac{1}{2}x$, where x is the amount of input used and y is the resulting amount of output, and the inequality reflects the fact that the firm could produce inefficiently, producing less output than the amount it *could* produce with the input amount x . The set of possible input-output combinations (vectors) for this firm would be $A = \{(x, y) \in \mathbb{R}_+^2 \mid y \leq \frac{1}{2}x\}$. Suppose a second firm's production possibilities are described by the production function $y \leq f_B(x) = \sqrt{x}$ with the inequality interpreted the same way. This firm's set of possible input-output combinations is $B = \{(x, y) \in \mathbb{R}_+^2 \mid y \leq \sqrt{x}\}$. Altogether, the aggregate input-output combinations (x, y) that are possible for the two firms are the ones in the set $A + B$ (again, assuming no externalities from one firm's production on the other firm's possibilities).

Example 6: Generally, we don't want to say ahead of time which goods are inputs and which are outputs. Which ones are which will depend on the firm, for example: a particular good might be an output for one firm and an input for another. If $x_k > 0$ for good k in an input-output plan $(x_1, x_2) \in \mathbb{R}^2$, then that's the amount of good k produced in the plan (x_1, x_2) ; and if $x_k < 0$, then the negative of the amount of good k used as input in the plan (x_1, x_2) . Then, $A = \{\mathbf{x} \in \mathbb{R}^2 \mid x_1 \leq 0, 0 \leq x_2 \leq \frac{1}{2}(-x_1)\}$ and $B = \{\mathbf{x} \in \mathbb{R}^2 \mid x_1 \leq 0, 0 \leq x_2 \leq \sqrt{-x_1}\}$.

Example 7: One firm's production possibilities set is $A = \{\mathbf{x} \in \mathbb{R}^2 \mid x_1 \leq 0, 0 \leq x_2 \leq \frac{3}{2}(-x_1)\}$ and the other firm's is $B = \{\mathbf{x} \in \mathbb{R}^2 \mid x_2 \leq 0, 0 \leq x_1 \leq \frac{1}{3}(-x_2)\}$. Note that $A + B$ includes the entire negative quadrant. For example, the plan $\mathbf{x}^A = (-4, 6)$ is in A and the plan

$\mathbf{x}^B = (3, -9)$ is in B , so the aggregate plan $\mathbf{x} = \mathbf{x}^A + \mathbf{x}^B = (-1, -3)$ is in $A + B$. Suppose the economy has an endowment $\mathbf{x} = (4, 9)$ of the two goods and allocates the endowment to the two firms so as to have Firm A do $\mathbf{x}^A$ and Firm B do $\mathbf{x}^B$. Now, the economy would have $\mathbf{x} + (-1, -3) = (3, 6)$ - i.e., less of both goods than it started with. That would be an especially bad way to allocate the endowment to production.

Exercise 1. Draw a diagram for Example 1.5, depicting the sets A , B , and $A + B$.

Theorem 1.3 (Sum-of-sets Maximization Theorem). Let $X_1, X_2, \dots, X_m$ be subsets of a vector space V , and for each $i = 1, \dots, m$ let $\bar{\mathbf{x}}_i = \sum_{i=1}^m X_i$ and let $f : V \rightarrow \mathbb{R}$ be a linear function. Then, $\bar{\mathbf{x}}$ maximizes f on X if and only if, for each $i = 1, \dots, m$, $\bar{\mathbf{x}}_i$ maximizes f on X_i .

Exercise 2. Prove Theorem 1.3.

Remark. In the vector space $\mathbb{R}^n$, real-valued linear functions have the form $f(\mathbf{x}) = \mathbf{p} \cdot \mathbf{x}$, so the theorem says that $\mathbf{x}$ maximizes $\mathbf{p} \cdot \mathbf{x}$ on X if and only if, for each $i = 1, \dots, m$, $\bar{\mathbf{x}}_i$ maximizes $\mathbf{p} \cdot \mathbf{x}$ on X_i . So, if $\mathbf{p}$ is a list of prices, then the theorem says that the “aggregate” vector $\bar{\mathbf{x}} = \sum_{i=1}^m \bar{\mathbf{x}}_i$ maximizes value on X if and only if each of the vectors $\bar{\mathbf{x}}_i$ maximizes value on X_i . For example, the aggregate production plan $\bar{\mathbf{x}}$ maximizes value (revenue or profit) on the set X of aggregate plans if and only if each $\bar{\mathbf{x}}_i$ maximizes value on the respective sets X_i . Because of this application of the theorem, it is often referred to as a “disaggregation” or “decentralization” theorem: it says that the decision about $\mathbf{x}$ can be decentralized or disaggregated, into separate decisions about the $\mathbf{x}_i$ vectors without compromising the objective of choosing a value-maximizing $\mathbf{x}$. Of course, this requires that f - i.e., value - be a linear function and that there are no external effects, in which the choice of some $\mathbf{x}_i$ affects some other set X_j of possible choices.

Exercise 3. Provide a counterexample to show that linearity of the function f is required in the theorem.

2 Sequences and Convergence of Sequences

Definition 2.1. A **sequence** in a set X (a sequence of elements of X) is a function $s : \mathbb{N} \rightarrow X$. We usually denote $s(n)$ by s_n , called the n -th term of s , and write $\{s_n\}$ for the sequence, or $\{s_1, s_2, \dots\}$.

Following the definition, a sequence of real numbers is an assignment of a real number to each natural number.

Definition 2.2. A sequence $\{x_n\}$ of real numbers **converges** to $\bar{x} \in \mathbb{R}$ if

$$\forall \varepsilon > 0 : \exists \bar{n} \in \mathbb{N} : n > \bar{n} \Rightarrow |x_n - \bar{x}| < \varepsilon$$

Example. The sequence $x_n = \frac{1}{n}$ converges to 0.

Example. The sequence $x_n = (-1)^n$ does not converge.

Example. The sequence

$$x_n = \begin{cases} 1, & \text{if } n \text{ is a square, i.e., if } n \in \{1, 4, 9, 16, \dots\} \\ 0, & \text{otherwise} \end{cases}$$

does not converge, despite the fact that it has ever longer and longer strings of terms that are zero.

If we replace $|x_n - \bar{x}|$ in this definition with the distance notation $d(x_n, \bar{x})$, then the definition applies to sequences in any metric space:

Definition 2.3. Let (X, d) be a metric space. A sequence $\{x_n\}$ in X **converges to** $\bar{x} \in X$ if

$$\forall \varepsilon > 0 : \exists \bar{n} \in \mathbb{N} : n > \bar{n} \Rightarrow d(x_n, \bar{x}) < \varepsilon.$$

We say that $\bar{x}$ is the limit of $\{x_n\}$, and we write $\lim\{x_n\} = \bar{x}$, $x_n \rightarrow \bar{x}$, and $\{x_n\} \rightarrow \bar{x}$.

Definition 2.4. Let $\{x_1, x_2, x_3, \dots\}$ be a sequence of real numbers and let r be a real number. r is **limit point** of this sequence if for any positive number ε , there is a positive integer N such that for all $n \geq N$, x_n is in the ε -interval about r ;

$$|x_n - r| < \varepsilon.$$

In this case, we say that the sequence **converges to** r and write as

$$\lim x_n = r \text{ or } \lim_{n \rightarrow \infty} x_n = r \text{ or } x_n \rightarrow r.$$

Limit points are also called **accumulation point** or **cluster point**.

Definition 2.5. Let (X, d) be a metric space, let $\bar{x} \in X$, and let $r \in \mathbb{R}_{++}$.

(1) The open ball about $\bar{x}$ of radius r is the set $B(\bar{x}, r) := \{x \in X \mid d(x, \bar{x}) < r\}$.

(2) The closed ball about $\bar{x}$ of radius r is the set $\bar{B}(\bar{x}, r) := \{x \in X \mid d(x, \bar{x}) \leq r\}$.

Remark. Let (X, d) be a metric space. A subset $S \subseteq X$ is bounded if and only if it is contained in an open ball - and equivalently, if and only if it is contained in a closed ball.

Remark. Let (X, d) be a metric space. A sequence $\{x_n\}$ in X converges to $\bar{x}$ if and only if for every $\varepsilon > 0$, " x_n is eventually in $B(\bar{x}, \varepsilon)$ " - i.e., $\exists \bar{n} \in \mathbb{N} : n > \bar{n} \Rightarrow x_n \in B(\bar{x}, \varepsilon)$.

Following proposition gives a proof that the sum of two convergent sequences in a normed vector space is a convergent sequence:

Proposition 2.6. *If $\{x_n\}$ and $\{y_n\}$ are sequences in a normed vector space $(V, \|\cdot\|)$ and if $\{x_n\} \rightarrow \bar{x}$ and $\{y_n\} \rightarrow \bar{y}$, then $\{x_n\} + \{y_n\} \rightarrow \bar{x} + \bar{y}$.*

Proof. Let $\varepsilon > 0$. Because $\{x_n\} \rightarrow \bar{x}$ and $\{y_n\} \rightarrow \bar{y}$, there are $\bar{n}_x$ and $\bar{n}_y$ such that

$$n > \bar{n}_x \Rightarrow \|x_n - \bar{x}\| < \varepsilon/2 \text{ and } n > \bar{n}_y \Rightarrow \|y_n - \bar{y}\| < \varepsilon/2.$$

Let $\bar{n} = \max\{\bar{n}_x, \bar{n}_y\}$, then

$$n > \bar{n} \Rightarrow \|x_n - \bar{x}\| + \|y_n - \bar{y}\| < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

But we have

$$\begin{aligned} \|(x_n + y_n) - (\bar{x} + \bar{y})\| &= \|(x_n - \bar{x}) + (y_n - \bar{y})\| \\ &\leq \|x_n - \bar{x}\| + \|y_n - \bar{y}\|, \text{ by the Triangle Inequality.} \end{aligned}$$

Therefore,

$$n > \bar{n} \Rightarrow \|(x_n + y_n) - (\bar{x} + \bar{y})\| < \varepsilon,$$

and since $x_n + y_n$ is the n -th term of the sequence $\{x_n\} + \{y_n\}$, this concludes the proof. $\square$

Definition 2.7. *Let $\{x_n\}$ be a sequence in X . A **subsequence** of $\{x_n\}$ is the sequence $\{x_{n_k}\}$ for a strictly increasing sequence $\{n_k\}$ in $\mathbb{N}$.*

Remark. A subsequence $\{x_{n_k}\}$ of a sequence $\{x_n\}$ in X is also a sequence in X .

Example. Let $x_n = n$ and $n_k = k^2$. Then, $x_n = \{1, 2, 3, 4, \dots\}$ and $\{x_{n_k}\} = \{1, 4, 9, 16, \dots\}$.

Example. Let $x_n = (-1)^n$, i.e., $\{x_n\} = \{-1, 1, -1, 1, \dots\}$.

If $n_k = 2k$, then $\{x_{n_k}\} = \{1, 1, 1, 1, \dots\}$.

If $n_k = 2k - 1$, then $\{x_{n_k}\} = \{-1, -1, -1, -1, \dots\}$.

If $n_k = k^2$, then $\{x_{n_k}\} = \{-1, 1, -1, 1, \dots\}$.

Example. Let $x_n = (-1)^{n \frac{n-1}{n}}$, i.e., $\{x_n\} = \{0, \frac{1}{2}, -\frac{2}{3}, \frac{3}{4}, -\frac{4}{5}, \dots\}$.

If $n_k = 2k$, then $\{x_{n_k}\} = \{\frac{1}{2}, \frac{3}{4}, \frac{5}{6}, \dots\}$.

If $n_k = 2k - 1$, then $\{x_{n_k}\} = \{0, -\frac{2}{3}, -\frac{4}{5}, -\frac{6}{7}, \dots\}$.

If $n_k = k^2$, then $\{x_{n_k}\} = \{0, \frac{3}{4}, -\frac{8}{9}, \frac{15}{16}, -\frac{24}{25}, \dots\}$.

3 Open and Closed Sets

Definition 3.1. Let $\mathbf{r}$ be a vector in $\mathbb{R}^n$ and let ϵ be a positive number. The ϵ -ball about $\mathbf{r}$ is

$$B_\epsilon(\mathbf{r}) \equiv \{\mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x} - \mathbf{r}\| < \epsilon\}.$$

In two-dimensional Cartesian plane, we can think of ϵ as a radius and $B_\epsilon(\mathbf{r})$ as a ball around $\mathbf{r}$ that has a radius of ϵ .

Definition 3.2. A set S in $\mathbb{R}^n$ is **open** if for each $\mathbf{x} \in S$, there exists an open ϵ -ball around $\mathbf{x}$ completely contained in S :

$$\mathbf{x} \in S \Rightarrow \text{there is an } \epsilon > 0 \text{ such that } B_\epsilon(\mathbf{x}) \subset S.$$

$B_\epsilon(\mathbf{x})$ is an open set, since we can always find another ϵ that satisfy the definition of “openness”. Intuitively, we can think “open” as “no boundary”. On the contrary, “closed” set would contain the boundaries.

Theorem 3.3. 1. $\emptyset$ and S are open sets.

2. If $S_1, S_2, \dots, S_n$ are open sets, then $\cap_{i=1}^n S_i$ is an open set.

3. Let A be an arbitrary set. If S_α is an open set for each $\alpha \in A$, then $\cup_{\alpha \in A} S_\alpha$ is an open set. In other words, the union of any collection of open sets is open. (Note that A can be any set, not necessarily, or even typically, a subset of X .)

Proof. 1. $\emptyset$ is open because it is vacuously true, i.e., there is no $x \in \emptyset$. X is open by the definition.

2. Let S_i be an open set for $i = 1, \dots, n$ and let $x \in \cap_{i=1}^n S_i$. We need to find an $\varepsilon > 0$ such that $B_\varepsilon(x) \subseteq \cap_{i=1}^n S_i$. For each i , there is ε_i such that $B_{\varepsilon_i}(x) \subseteq S_i$. Let $\varepsilon = \min\{\varepsilon_1, \dots, \varepsilon_n\}$. Then, $0 < \varepsilon \leq \varepsilon_i$ for each i and $B_\varepsilon(x) \subseteq B_{\varepsilon_i}(x) \subseteq S_i$ for each i , i.e., $B_\varepsilon(x) \subseteq \cap_{i=1}^n S_i$.
3. Let $x \in \cup_{\alpha \in A} S_\alpha$. We need to find $\varepsilon > 0$ for which $B_\varepsilon(x) \subseteq \cup_{\alpha \in A} S_\alpha$. Since $x \in \cup_{\alpha \in A} S_\alpha$, we have $x \in S_{\bar{\alpha}}$ for some $\bar{\alpha} \in A$. Since $S_{\bar{\alpha}}$ is open, there is an $\varepsilon > 0$ such that $B_\varepsilon(x) \subseteq S_{\bar{\alpha}} \subseteq \cup_{\alpha \in A} S_\alpha$.

□

Definition 3.4. A set S is **closed** if it is the complement of an open set.

Theorem 3.5. 1. $\emptyset$ and X are closed sets.

2. If $S_1, S_2, \dots, S_n$ are closed set, then $\cup_{i=1}^n S_i$ is a closed set.

3. Let A be an arbitrary set. If S_α is a closed set for each $\alpha \in A$, then $\cap_{\alpha \in A} S_\alpha$ is a closed set.

Exercise 4. Prove *Theorem 3.5-2* and *Theorem 3.5-3* using De Morgan's Laws.

In the following theorem, limit points provide an important characterization of closed sets. As always with characterizations, this characterization is an alternative definition of a closed sets. In fact, many people actually use this as the definition of a closed set, and then the definition we are using, given above, becomes a theorem that provides a characterization of closed sets as complements of open sets.

Theorem 3.6. A set is closed if and only if it contains all its limit points.

Proof. ($\Rightarrow$): Let S be a closed set and let $\{x_n\}$ be a sequence in S (i.e., $\forall n \in \mathbb{N} : x_n \in S$) that converges to $\bar{x} \in X$. We must show that $\bar{x} \in S$. Suppose not, i.e., $\bar{x} \in S^c$. Since S^c is open, there is an $\varepsilon > 0$ for which $B_\varepsilon(\bar{x}) \subseteq S^c$. Since $\{x_n\} \rightarrow \bar{x}$, let $\bar{n}$ be such that $n > \bar{n} \Rightarrow |x_n - \bar{x}| < \varepsilon$. Then, for $n > \bar{n}$ we have both $x_n \in S$ and $x_n \in B_\varepsilon(\bar{x}) \subseteq S^c$, a contradiction.

($\Leftarrow$): Suppose S is not closed. We must show that S does not contain all its limit points. Since S is not closed, S^c is not open. Therefore, there is at least one element $\bar{x}$ of S^c such that every ball $B_\varepsilon(\bar{x})$ contains at least one element of $(S^c)^c = S$. For every $n \in \mathbb{N}$, let $x_n \in B_{\frac{1}{n}}(\bar{x}) \cap S$. Then, we have a sequence $\{x_n\}$ in S which converges to $\bar{x} \notin S$, i.e., $\bar{x}$ is a limit point of S but is not in S , so S does not contain all its limit points. □

Example 1: For each $n \in \mathbb{N}$, let S_n be an open set $(-\frac{1}{n}, \frac{1}{n}) \subseteq \mathbb{R}$. Then, $\cap_{n=1}^{\infty} S_n = \{0\}$, which is not open. This is a counterexample which shows that **Theorem 3.3-2** would not necessarily hold if the collection weren't finite.

Example 2: For each $n \in \mathbb{N}$, let S_n be a closed set $[\frac{1}{n}, \frac{n-1}{n}] \subseteq \mathbb{R}$. Then, $\cup_{n=1}^{\infty} S_n = (0, 1)$, which is not closed. This is a counterexample which shows that **Theorem 3.5-2** would not necessarily hold if the collection weren't finite.

Example 3: $\mathbb{R}_{++}^n$ is open. This can be shown directly, by finding an appropriate $\varepsilon > 0$ for each $x \in \mathbb{R}$. Alternatively, one could show that for each $i = 1, \dots, n$ the set $S_i = \{\mathbf{x} \in \mathbb{R}^n | x_i > 0\}$ is open and invoke **Theorem 3.3-2** for the set $\mathbb{R}_{++}^n = \cap_{i=1}^n S_i$.

Example 4: The union of all open subsets of $\mathbb{R}_+^n$ is an open set, according to **Theorem 3.3-3**. Note that this set is $\mathbb{R}_{++}^n$. This is another way to show that $\mathbb{R}_{++}^n$ is an open set.

Example 5: Let G be any subset of (X, d) and let $\dot{G}$ be the union of all open subsets of G . According to **Theorem 3.3-3**, $\dot{G}$ is an open set. $\dot{G}$ is clearly the “largest” open subset of G , in the sense that (i) $\dot{G}$ is itself an open subset of G , and (ii) every open subset of G is a subset of $\dot{G}$, i.e., $\dot{G}$ contains every open subset of G , which we could state informally as $\dot{G}$ is “at least as large” as every other open subset of G . We call the set $\dot{G}$ the **interior** of G , also denoted as $\text{int } G$.

Example 6: Let G be any subset of (X, d) and let $\bar{G}$ be the intersection of all closed sets that contain G . According to **Theorem 3.5-3**, $\bar{G}$ is a closed set. It is the “smallest” closed set containing G as a subset, in the sense that (i) $\bar{G}$ is itself a closed set containing G , and (ii) every closed set containing G as a subset also contains $\bar{G}$ as a subset. We could state that every other closed set containing G is “at least as large” as $\bar{G}$. We call $\bar{G}$ the **closure** of G , also denoted as $\text{cl } G$.

Definition 3.7. Let G be a subset of (X, d) . The **interior** of G , denoted $\text{int } G$ or $\dot{G}$, is the union of all open subsets of G , and the **closure** of G , denoted $\text{cl } G$ or $\bar{G}$, is the intersection of all closed sets that contain G .

Remark. $\dot{G} \subseteq G \subseteq \bar{G}$, i.e., $\text{int } G \subseteq G \subseteq \text{cl } G$.

The interior of the complement of G , i.e., $\text{int } G^c$, is called the *exterior* of G .

Definition 3.8. Let G be a subset of (X, d) . The **exterior** of G , denoted $\text{ext } G$, is the interior of G^c .

Remark. The exterior of G is the union of all open sets that do not intersect G , i.e., the largest open set in G^c .

The *boundary* of a set lies “between” its interior and exterior:

Definition 3.9. Let G be a subset of (X, d) . The **boundary** of G , denoted $\text{bdy } G$, is the complement of $\text{int } G \cup \text{ext } G$, i.e., $\text{bdy } G = [\text{int } G \cup \text{ext } G]^c$.

Remark. The interior, exterior, and boundary of a set comprise a partition of the set. The interior and exterior are both open, and the boundary is closed. $\text{bdy } G = \text{cl } G \cap \text{cl } G^c$.

Example 7: Let $u : \mathbb{R}_{++}^n \rightarrow \mathbb{R}$ be defined by $u(x_1, x_2) = x_1 x_2$ and let $S = \{\mathbf{x} \in \mathbb{R}_{++}^2 \mid u(\mathbf{x}) < \xi\}$ for some $\xi \in \mathbb{R}_{++}$. Note that $S = u^{-1}((-\infty, \xi))$, the inverse image under u of the open interval $(-\infty, \xi)$, in fact, S is the inverse image of the open interval (c, ξ) for any $c \leq 0$, and S is also the lower-contour set of any $\bar{\mathbf{x}}$ such that $u(\bar{\mathbf{x}}) = \xi$. We will show that S is an open set.

Proof. Let $\bar{\mathbf{x}} \in S$, i.e., $\bar{x}_1 \bar{x}_2 < \xi$. We must find $\varepsilon > 0$ such that $B_\varepsilon(\bar{\mathbf{x}}) \subseteq S$, i.e., such that

$$\mathbf{x} \in B_\varepsilon(\bar{\mathbf{x}}) \Rightarrow x_1 x_2 < \xi \quad (1)$$

For whatever value of ε we choose, we have

$$\begin{aligned} \mathbf{x} \in B_\varepsilon(\bar{\mathbf{x}}) &\Rightarrow [x_1 < \bar{x}_1 + \varepsilon \ \& \ x_2 < \bar{x}_2 + \varepsilon] \\ &\Rightarrow x_1 x_2 < (\bar{x}_1 + \varepsilon)(\bar{x}_2 + \varepsilon) \\ &\Leftrightarrow x_1 x_2 < \bar{x}_1 \bar{x}_2 + (\bar{x}_1 + \bar{x}_2)\varepsilon + \varepsilon^2 \end{aligned}$$

Thus, if we set $\bar{x}_1 \bar{x}_2 + (\bar{x}_1 + \bar{x}_2)\varepsilon + \varepsilon^2 = \xi$, we satisfy [Eq. \(1\)](#). Solving for ε , we have

$$\varepsilon = \frac{1}{2} \left[-(\bar{x}_1 + \bar{x}_2) + \sqrt{(\bar{x}_1 + \bar{x}_2)^2 + 4(\xi - \bar{x}_1 \bar{x}_2)} \right] > 0$$

□

Example 7 is an important example, not the specific utility function $u(x_1, x_2) = x_1 x_2$, but the fact that the set S we defined (the inverse image under u of an open set in $\mathbb{R}$) was itself an open set in the domain of u . In fact, it is a theorem that a function $f : (X, d) \rightarrow (Y, d')$ is continuous if and only if $f^{-1}(A)$ is open in X for every open set $A \subseteq Y$. These notion of continuity, and their relationship to open sets, have important implications for utility theory and other applications in economics.

4 Bolzano-Weierstrass Theorem

This section presents some properties of sequences that are important in economic analysis. We first study the definition of compact set that builds an important theorem, Bolzano-Weierstrass Theorem. The compact sets have two properties that give them an important role in economic theory: (1) every sequence in a compact set has a subsequence that converges, and (2) every continuous function on a compact set attains the maximum and minimum on its compact domain. Most economics theorems that assure the existence of an economic equilibrium rely on these two results. Before we state Bolzano-Weierstrass Theorem, first we build some definitions to prove Bolzano-Weierstrass Theorem.

4.1 Cauchy Sequences

Definition 4.1. A sequence $\{x_n\}$ of real numbers is **bounded** if there is a number $M \in \mathbb{R}$ for which every term x_n satisfies $|x_n| \leq M$.

Definition 4.2. A set S in $\mathbb{R}^n$ is **compact** if and only if it is both closed and bounded.

So far, we only talk about the convergence of a sequence using a limit. However, we can discuss about the convergence without actually finding a limit and the idea is simply based on that the elements should get arbitrarily close to each other, if the elements of a sequence is getting close to some limit.

Definition 4.3. A sequence $\{x_n\}_{n=1}^{\infty}$ is a **Cauchy sequence** if for any positive number ε , there is an integer N such that, for all $i, j \geq N$, $d(x_i, x_j) < \varepsilon$.

Theorem 4.4. Any convergent sequence in $\mathbb{R}^n$ is Cauchy sequence.

Proof. Suppose that $\{x_n\}_{n=1}^{\infty}$ converges to x . Given $\varepsilon > 0$, choose N such that $d(x_n, x) < \varepsilon/2$ for all $n \geq N$. Then, for $n, m \geq N$,

$$d(x_n, x_m) \leq d(x_n, x) + d(x_m, x) < \varepsilon$$

by triangle inequality. □

Definition 4.5. Let S be a subset of $\mathbb{R}$. An **upper bound** of S is a number b such that $x \leq b$ for every $x \in S$. A **least upper bound** of S is a b^* such that $b^* \leq b$ for every b that is an upper bound of S .

Remark. In $\mathbb{R}$, a set can have no more than one least upper bound, so it makes sense to talk about the least upper bound of S . It is also called the *supremum* of S , denoted $\sup S$ or $\text{lub } S$.

Definition 4.6. A sequence $\{x_n\}_{n=1}^{\infty}$ of real numbers is **monotone increasing** if each entry in the sequence is at least as big as the previous entry, i.e., $x_{n-1} \leq x_n$ for all n . It is **monotone decreasing** if $x_n \leq x_{n-1}$ for all n . It is **monotone** if it is either monotone increasing or monotone decreasing. Such a sequence is **strictly monotone increasing (decreasing)** if $x_{n-1} < x_n$ ($x_n < x_{n-1}$) for all n .

Most sequences, of course, do not converge. Even if we restrict attention to bounded sequences, there is no reason to expect that a bounded sequence converges. The following theorem gives us condition that is sufficient to ensure that a sequence converges.

Theorem 4.7 (The Monotone Convergence Theorem:). Every bounded monotone sequence in $\mathbb{R}$ converges to an element of $\mathbb{R}$.

Proof. Let $\{x_n\}$ be a monotone increasing sequence of real numbers. Since it is bounded, it has a least upper bound b . We will show that $\{x_n\} \rightarrow b$. Suppose $\{x_n\}$ doesn't converge to b . Then, for some $\varepsilon > 0$, infinitely many terms of the sequence satisfy $|x_n - b| \geq \varepsilon$, i.e., $x_n \leq b - \varepsilon$ (x_n cannot be greater than b if b is an upper bound). It follows that $x_n \leq b - \varepsilon$ for all $n \in \mathbb{N}$: since $\{x_n\}$ is increasing, if $x_m > b - \varepsilon$ for some m , then $x_n > b - \varepsilon$ for all $n > m$, contradicting that $x_n \leq b - \varepsilon$ for infinitely many n . Thus, we have $x_n \leq b - \varepsilon$ for all $n \in \mathbb{N}$; i.e., $b - \varepsilon$ is an upper bound of $\{x_1, x_2, \dots\}$, and therefore b is not a least upper bound of $\{x_1, x_2, \dots\}$, a contradiction. Therefore, $\{x_n\}$ does converge to b .

If $\{x_n\}$ is a monotone decreasing sequence, the above proof shows that the increasing sequence $\{-x_n\}$ converges, and therefore $\{x_n\}$ converges. $\square$

We can also prove that all Cauchy sequences in $\mathbb{R}^n$ converges by proving the every Cauchy sequence is bounded and every sequence has a monotone subsequence. After proving these two parts, [Theorem 4.7](#) proves that all Cauchy sequences are convergent. We skip the proof for this. (If you are interested, check Theorem 29.2, Lemma 29.1, Lemma 29.2, Lemma 29.3 and Theorem 29.3 in SB-page 805.)

4.2 Bolzano-Weierstrass Theorem

Theorem 4.8. Any sequence contained in the closed and bounded interval $[0, 1]$ has a convergent subsequence.

Proof. Divide $[0, 1]$ into two subintervals, $[0, \frac{1}{2}]$ and $[\frac{1}{2}, 1]$. At least one of the two intervals contains infinitely many terms of $\{x_n\}$; denote that the interval by I_1 , which has length $\frac{1}{2}$ and let $\{x_{n'}\}$ be the subsequence of $\{x_n\}$ consisting of every term that lies in I_1 .

Now divide I_1 into two halves, each of length $\frac{1}{4}$, at least one of which contains infinitely many terms of the (sub)sequence $\{x_{n'}\}$ and denote that half by I_2 . Let $\{x_{n''}\}$ be the subsequence of $\{x_{n'}\}$ consisting of all of the terms that lie in I_2 . Continuing in this way, we construct a sequence of nested intervals $I_1 \supseteq I_2 \supseteq \cdots$, where the length of I_n is $\left(\frac{1}{2}\right)^n$, and each interval contains an infinite number of terms of the original sequence $\{x_n\}$. Finally, we construct a subsequence $\{z_n\}$ of $\{x_n\}$ made up of one term from each interval I_n . This subsequence is Cauchy sequence: for $m, n > N \Rightarrow |z_m - z_n| < \left(\frac{1}{2}\right)^N$. $\square$

Theorem 4.9. *Any sequence contained in a closed and bounded subset of $\mathbb{R}^n$ has a convergent subsequence.*